

Chapter 5: Elasticity and Its Application

The Price Elasticity of Demand

- **Elasticity** is a measure of the **responsiveness** of quantity demanded or quantity supplied to a change in one of its determinants.
- The **price elasticity of demand** measures how much the **quantity demanded** of a good responds to a **change in its price**
- Demand is **elastic** if quantity demanded responds **substantially** to a price change
- Demand is **inelastic** if quantity demanded responds only **slightly** to a price change
- Demand is **unit elastic** when the elasticity is exactly 1.

Factors that Determine the Price Elasticity of Demand

- Availability of close substitutes
- Necessities vs luxuries
- Definition of the market
- Time horizon

Mathematical Formulae

Formula for Price Elasticity of Demand

$$\text{Price elasticity of demand} = \left| \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}} \right|$$

Midpoint Method for Price Elasticity of Demand

Given the initial price P_1 , the initial quantity Q_1 , the final price P_2 , and the final quantity Q_2 , we have:

$$\text{Price elasticity of demand} = \left| \frac{(Q_2 - Q_1) \div \frac{Q_2 + Q_1}{2}}{(P_2 - P_1) \div \frac{P_2 + P_1}{2}} \right|$$

In general, the “midpoint method” simply refers to using the midpoint as the denominator (even for e.g. percentage change in price or quantity).

Total Revenue

- **Total revenue** is the amount sellers receive from sales: the price received per unit times the quantity sold $P \times Q$. Without a *tax wedge*, this also equals *buyers' total spending*.

The Elasticity–Revenue Rule

Along a fixed demand curve, **elasticity** determines whether the **price effect** or the **quantity effect** on total revenue, $P \times Q$, is **stronger**.

Demand	Price rises	Price falls
Inelastic ($E < 1$)	Revenue rises	Revenue falls
Unit elastic ($E = 1$)	Revenue unchanged	Revenue unchanged
Elastic ($E > 1$)	Revenue falls	Revenue rises

Summary

With inelastic demand, price and revenue move in the same direction. With elastic demand, they move in opposite directions.

Other Demand Elasticities

- The **income elasticity of demand** measures how much the quantity demanded of a good responds to a change in **consumer's income**. It is the percentage change in quantity demanded divided by the percentage change in income.
 - Here, keep the sign. **Normal goods would be positive, inferior goods would be negative**
- The **cross-price elasticity of demand** is the percentage change in quantity demanded of **one good** divided by the percentage change in price of **another good**
 - Here, keep the sign. **Substitutes would be positive, complements would be negative**

The Price Elasticity of Supply

- The **price elasticity of supply** measures how much the **quantity supplied** of a good **responds to a change in its price**. It is the percentage in quantity supplied divided by the percentage change in price.
- Supply is **elastic** if quantity supplied responds **substantially** to price
- Supply is **inelastic** if quantity supplied responds **only slightly** to price

Applications

- Interdiction** refers to efforts to **restrict supply**, such as intercepting shipments and arresting smugglers.